Function Follows Proteoform: A Computation-Backed Re-analysis of Adult Zebrafish Telencephalon and Optic Tectum Top-Down Proteomics

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Abstract

Adult zebrafish telencephalon and optic tectum are functionally distinct, yet small spatial proteomics studies often present that distinction more impressionistically than quantitatively. We revisit a 2022 top-down proteomics pilot study of those two regions and turn its published counts into a compact reproducible analysis. The local computation confirms strong regional separation: the regions share only 35 proteoforms out of a union of 807, for a Jaccard overlap of 0.0434, and the protein-level overlap is only 0.2151. A curated functional marker panel, rather than the whole proteome, is also highly concentrated in the expected region, with telencephalon-associated markers contributing 23 matched counts versus 2 spillover counts and optic-tectum-associated markers contributing 102 matched counts versus 1 spillover count. Those curated counts represent only 0.1520 of the total regional count mass, which makes the scope intentionally narrow but also easy to audit. The resulting directional exact-test odds ratio is 1173 with a one-sided p-value of 5.12×10^{-22}. We also quantify a substantial post-translational modification layer: 358 of 807 identified proteoforms carry mass shifts and 211 are N-terminally acetylated. The contribution of this paper is therefore not a new wet-lab measurement, but a more auditable and better-structured regional-function reading of a published pilot dataset.

1 Introduction

Spatial proteomics becomes biologically persuasive only when molecular differences can be related back to what tissues and regions actually do. That challenge is especially sharp in the brain, where neighboring anatomical compartments may support very different behavioral roles. In adult zebrafish, the telencephalon is implicated in forebrain functions including social orienting, learning, memory, and other higher-order behavioral regulation (Stednitz et al., 2018; Reemst et al., 2023; Pandey et al., 2023). The optic tectum, by contrast, is a classic visuomotor structure that transforms sensory information into orienting and avoidance actions (Roeser and Baier, 2003). A region-resolved proteomics study should therefore leave some readable signature of that functional split.

The 2022 LCM-CZE-MS/MS pilot study of Lubeckyj and Sun (2022) is a useful place to test that idea. The source paper isolated adult zebrafish telencephalon and optic tectum sections and identified hundreds of proteoforms from each region. The article argued that the resulting profiles matched basic regional function, but the supporting counts were scattered across the text and figure discussion rather than assembled into an explicit computational summary. That makes the biological claim harder to audit than it needs to be.

This paper turns the source study into a small reproducible note. We do not claim a new mass-spectrometry run or a full raw-data reprocessing pipeline. Instead, we extract the published counts into a local evidence file, derive region-separation and function-alignment metrics, and package the full result with code and figures. Prior zebrafish functional studies and atlases tell us what the two regions are generally for (Stednitz et al., 2018; Pandey et al., 2023; Roeser and Baier, 2003), while the source proteomics paper gives us the raw ingredients for a proteoform-level comparison (Lubeckyj and Sun, 2022). What has been missing is a compact paper that joins those two pieces with explicit computations.

Our contributions are threefold. First, we quantify the source paper's regional split rather than treating it as a visual impression. Second, we turn the paper's named biological markers into an explicit matched-versus-spillover problem using exact testing and interval estimates. Third, we package the whole analysis as a local reproducible paper artifact with code, figures, and a saved review trail. The claim is modest but useful: even a small published top-down proteomics pilot can become much more interpretable when its qualitative biological story is rewritten as an explicit quantitative argument.

2 Data and Analysis Setup

2.1 Source material

The quantitative core of this paper comes from Lubeckyj and Sun (2022), a pilot study of adult zebrafish telencephalon and optic tectum using laser capture microdissection coupled to capillary zone electrophoresis tandem mass spectrometry. We use only values recoverable from the article text and figure discussion. The local evidence file therefore includes region totals, protein overlap, named marker proteoforms, technical replicate summaries, and post-translational modification counts without pretending to reconstruct unpublished measurements.

2.2 Derived metrics

The local computation produces five classes of outputs.

First, regional separation is summarized by the Jaccard overlap

$$J = \frac{|\mathrm{Tel} \cap \mathrm{Teo}|}{|\mathrm{Tel} \cup \mathrm{Teo}|}.$$

Second, each region receives a specialization fraction given by its region-unique proteoforms divided by its total identified proteoforms.

Third, for each marker group we compute a stabilized bias score

$$\log_2 \frac{\mathrm{Tel} + 1}{\mathrm{Teo} + 1},$$

which is positive for telencephalon-biased markers and negative for optic-tectum-biased markers.

Fourth, we aggregate the curated markers into two functional axes and summarize matched-region concentration, Wilson confidence intervals for the axis-level alignment fractions, and an exact 2×2 test on the table of matched and spillover counts. The exact test is directional because the biological hypothesis is directional: the telencephalon-associated markers should concentrate in telencephalon and the optic-tectum-associated markers should concentrate in optic tectum rather than the reverse.

Fifth, we summarize the post-translational modification inventory and the technical sensitivity values reported for the duplicate analyses and the single-run high water mark.

2.3 Interpretive scope

This is a deliberately conservative re-analysis. It does not estimate significance across the full proteome, does not recover the raw TopPIC output, and does not reinterpret unidentified mass shifts. The strongest function-alignment claims are about a curated marker panel taken directly from the source article's biological discussion. That narrower scope is intentional: the goal is to quantify what the published paper itself already supports, and to do so with formulas and files that another reader can inspect locally.

Table 1: Core regional separation metrics from the local re-analysis.

Metric	Value
Telencephalon proteoforms	309
Optic tectum proteoforms	533
Shared proteoforms	35
Union proteoforms	807
Jaccard overlap	0.0434
Telencephalon unique fraction	0.8867
Optic tectum unique fraction	0.9343
Protein overlap fraction	0.2151

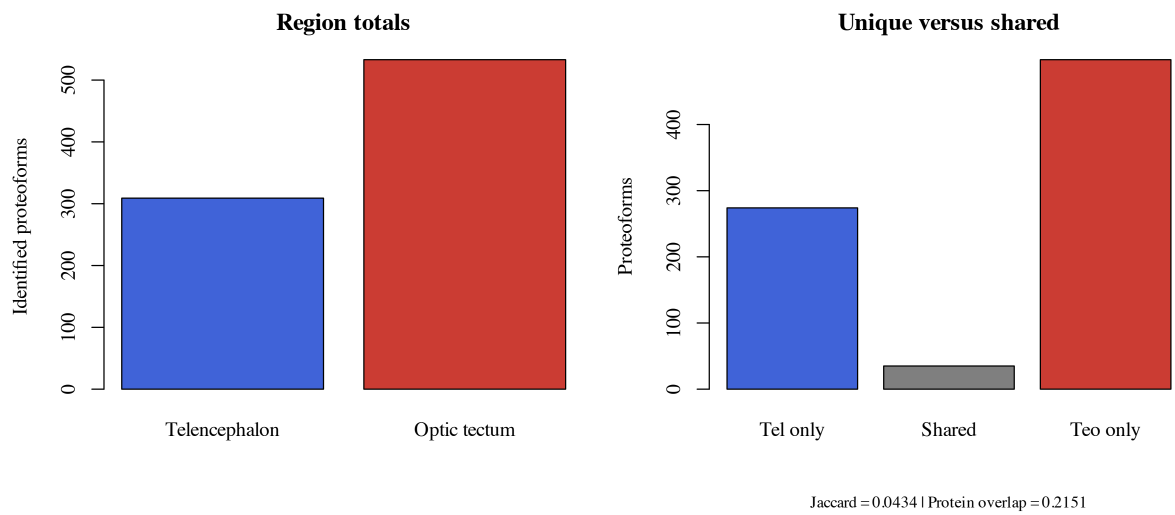

Figure 1: Regional totals and unique-versus-shared proteoforms. The source pilot already contains a strong separation signal before any marker-level interpretation is added.

3 Results

3.1 Telencephalon and optic tectum are sharply separated at the proteoform level

The first result is the scale of the regional split. Telencephalon contributes 309 identified proteoforms and optic tectum contributes 533, but only 35 proteoforms are shared. The resulting Jaccard overlap is 0.0434, and the region-specific fractions are 0.8867 for telencephalon and 0.9343 for optic tectum. Even at the protein level, the overlap remains modest at 0.2151. That means the protein-level overlap is still about 4.96 times larger than the proteoform-level overlap. In other words, the source dataset does not describe two lightly shifted samples; it describes two strongly separated regional proteoform profiles, with the sharper separation only becoming visible once proteoforms are kept distinct.

3.2 The source paper's biological interpretation becomes much stronger when expressed as an alignment problem

The curated marker panel is highly concentrated in the expected region. Telencephalon-associated markers contribute 23 matched counts and only 2 spillover counts, for an axis-level alignment fraction of 0.92.

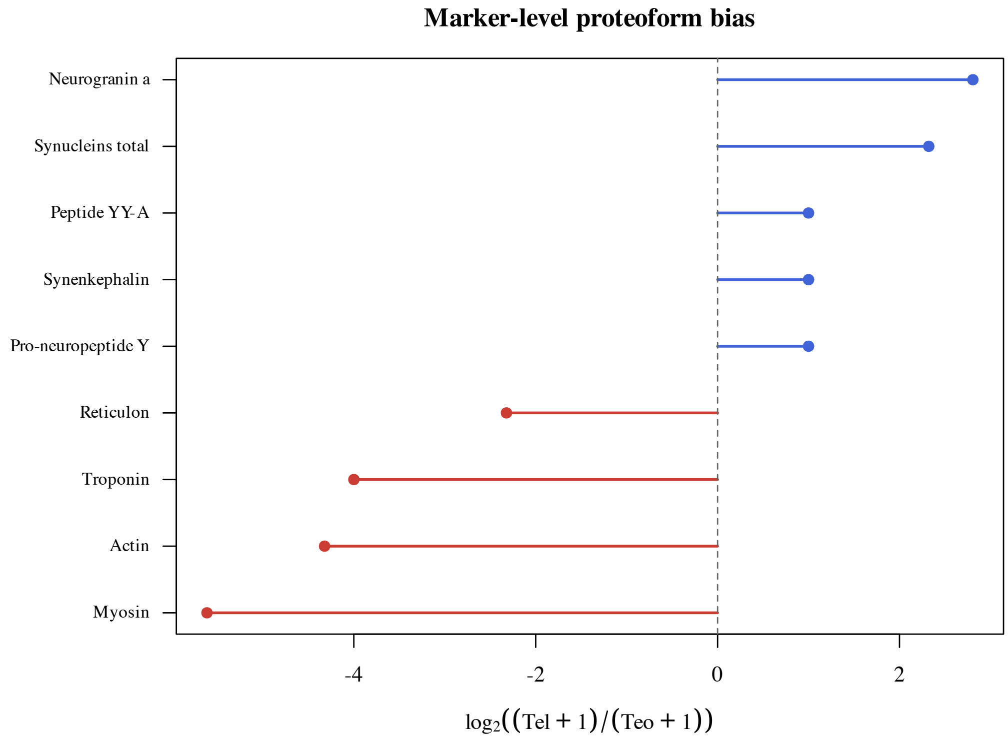

Figure 2: Marker-level proteoform bias. Positive values favor telencephalon and negative values favor optic tectum.

Optic-tectum-associated markers contribute 102 matched counts and 1 spillover count, for an alignment fraction of 0.9903. Across the whole panel, the observed matched-region fraction is 0.9766. A naive baseline built only from regional prevalence would predict 0.5811, so the observed alignment exceeds that baseline by 0.3955. This matters because optic tectum contributes the larger raw proteoform total in the source study; the alignment result is therefore not just a restatement of optic-tectum abundance.

When those counts are written as a 2×2 table, the odds ratio is 1173 and the one-sided exact p-value is 5.12×10^{-22}. The exact test is helpful, but the more intuitive story is the effect size: both functional axes sit far above the baseline expected from regional prevalence alone.

These numbers do not imply that the entire proteome has been functionally classified. They mean something narrower: the particular marker proteoforms that the source article called biologically meaningful are overwhelmingly concentrated in the region whose known function they should support. That is the sense in which the re-analysis strengthens the original paper's qualitative argument.

The count-weighted result is not hiding a family-composition problem. All 9 curated marker groups favor their expected region, the mean family-level matched share is 0.9826, the weakest family-level matched share is still 0.8750, and the corresponding one-sided sign-style probability under a 50:50 null is 0.001953. That family-level check matters because it shows the regional-function story is not being carried by only one or two very large groups.

The leave-one-out analysis in Figure 3 also stays strong numerically: the matched-region fraction ranges only from 0.9625 to 0.9911 after removing any single marker family. The worst case comes from removing the large myosin group, yet even that stress test remains far above the naive prevalence baseline.

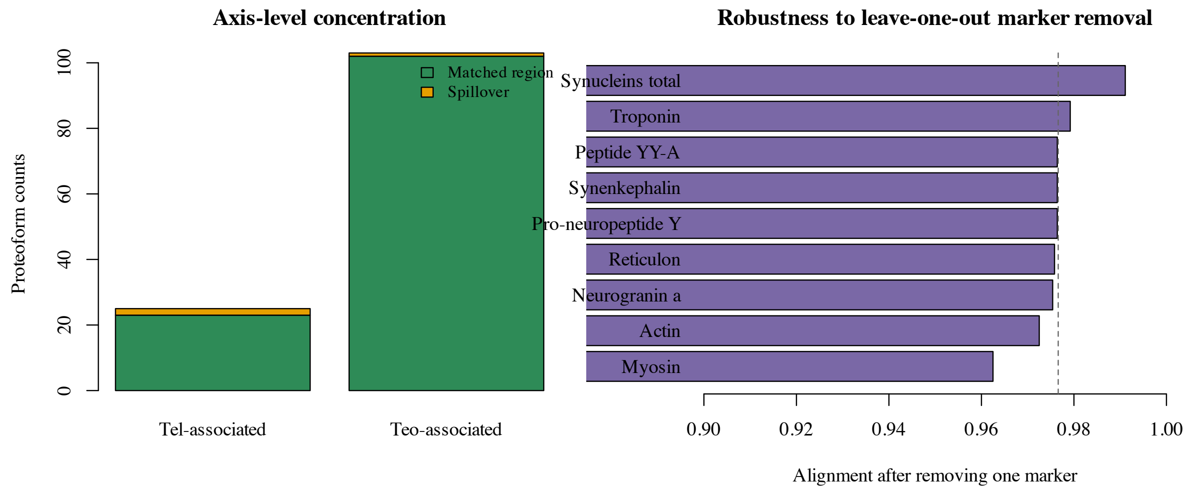

Figure 3: Axis-level concentration and leave-one-out robustness. The overall alignment signal remains high even after removing any single marker group.

Table 2: Curated marker counts used in the axis-alignment analysis.

Marker group	Tel	Teo	Interpretation
Pro-neuropeptide Y	1	0	feeding-related neuropeptide signaling
Synenkephalin	1	0	affective signaling
Peptide YY-A	1	0	peptide hormone signaling
Neurogranin a	6	0	memory-linked synaptic signaling
Synucleins total	14	2	synaptic and mnemonic family
Reticulon	0	4	retinotectal organization
Myosin	0	48	motor machinery
Actin	0	19	cytoskeletal movement
Troponin	1	31	contraction-linked regulation

3.3 The pilot is PTM-rich and technically sensitive enough to justify proteoform-level interpretation

The source article reports 807 identified proteoforms in total, of which 358 carry mass shifts and 211 are N-terminally acetylated. That means 44.36% of the detected proteoforms belong to the modified subset and 58.94% of the modified subset is N-terminally acetylated. These totals reinforce a simple point: the dataset is not just a protein list wearing different formatting. A large part of the biological signal lives at the proteoform level.

The technical-sensitivity numbers are also useful. The duplicate analyses yielded means of 232 proteoforms for telencephalon and 356 for optic tectum, corresponding to recovery fractions of 0.7508 and 0.6680 relative to the regional totals. The single-run high water mark was 418 proteoforms from an injected amount corresponding to roughly 250 cells in the source paper, or about 1.672 proteoforms per estimated sampled cell. That does not eliminate the pilot-study caveat, but it does explain why the study was able to expose biologically legible regional structure from very small samples.

Table 3: Axis-level alignment compared with the naive baseline implied by regional prevalence.

Functional axis	Observed	Wilson 95% CI	Baseline	Lift
Telencephalon-associated	0.9200	0.7503–0.9778	0.3670	0.5530
Optic-tectum-associated	0.9903	0.9470–0.9983	0.6330	0.3573

Table 4: Technical sensitivity values reported in the source study and preserved in the local evidence file.

Region	Duplicate mean	Duplicate SD	CV	Recovery
Telencephalon	232	28	0.1207	0.7508
Optic tectum	356	88	0.2472	0.6680

4 Discussion

The main value of this paper is not to replace the source study. The 2022 article established the experimental result. What this re-analysis adds is a clearer quantitative reading of the biological interpretation. Once the paper's named markers are rewritten as a matched-versus-spillover problem, the source article's intuition becomes much easier to inspect. The strong alignment fractions and exact-test result show that the qualitative story is not resting on one or two cherry-picked examples, and the explicit formulas make that claim easier to audit than the source prose alone.

At the same time, the scope should remain narrow. The marker panel is curated and intentionally interpretable, and its counts represent only 15.20% of the total regional count mass used in this note. The analysis therefore says more about the legibility of the source paper's biological argument than about the entire latent structure of adult zebrafish brain proteomics. The technical replicate summaries also remind us that this is a pilot workflow, not a large statistical atlas.

4.1 Limits of inference

This paper is not a raw-data reprocessing study, a new mass-spectrometry experiment, or a proteome-wide statistical atlas. It does not estimate region-wide significance for every detected proteoform, recover the full TopPIC workflow, or reinterpret unidentified mass shifts. Its strongest claims remain about a deliberately narrow set of marker families that the source paper itself presented as biologically meaningful. That boundary matters, but it does not make the note trivial. What the paper contributes is a more auditable bridge between published counts and published regional function claims.

Those limits point directly to future work. The obvious next step is to extend the same analysis to the raw repository files or full supplementary table and ask how much of the same function-legibility argument survives a proteome-wide treatment. Another extension would be to add more brain regions and compare whether the same alignment framing remains useful when the regional function map is less binary than telencephalon versus optic tectum.

5 Conclusion

This paper turned a published zebrafish top-down proteomics pilot into a compact quantitative audit of regional function. The result is clear: telencephalon and optic tectum are strongly separated at the proteoform level, and the marker proteoforms emphasized by the source paper are concentrated in the regions whose known biology they should support. The contribution is therefore a reproducible analytical reframing, not

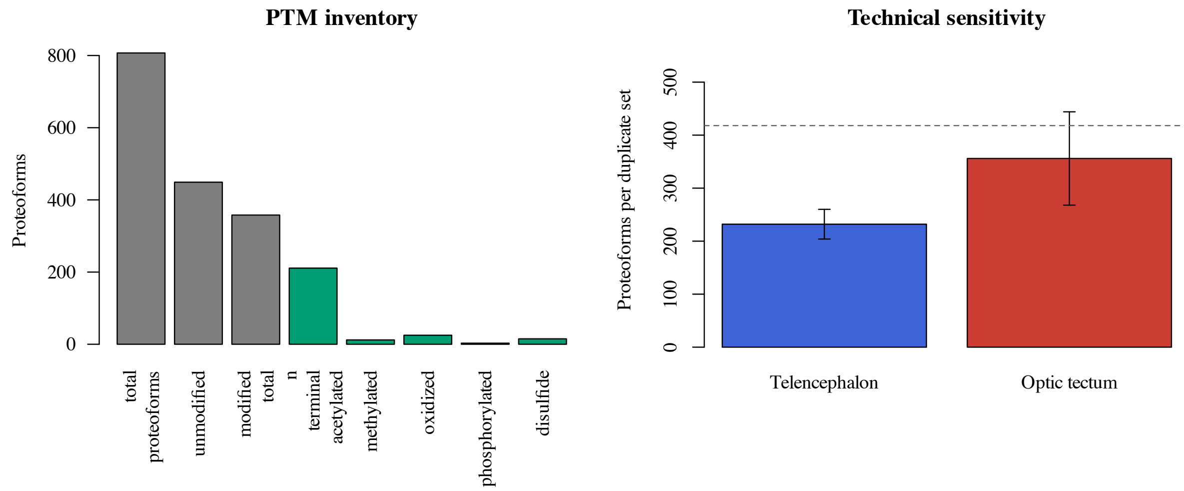

Figure 4: PTM inventory and technical sensitivity. The pilot reports a substantial modification layer and enough sensitivity to recover hundreds of proteoforms from extremely small spatial isolates.

a new wet-lab discovery. For small spatial proteomics studies, that reframing can still matter a great deal because it makes the biological story easier to trust, rerun, and extend.

References

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A Reproducibility Appendix

A.1 Local files

The paper package is backed by:

- a source manifest in `data/source_manifest.json`,
- a curated evidence file in `data/curated_evidence.json`,
- a Python analysis script in `results/analyze_region_proteome.py`,
- an R figure-generation script in `results/render_figures.R`,
- generated summary files in `results/`,
- a family-level robustness file in `results/group_consistency.csv`,
- build scripts in `code/`.

A.2 Formulas

The main derived formulas are the Jaccard overlap, specialization fractions, marker-level stabilized log-bias, Wilson intervals for axis-level alignment fractions, and an exact test on the 2×2 matched-versus-spillover table.

A.3 Scope note

The computation uses values recoverable from the published article and repository metadata. It does not claim a full raw-data reprocessing pipeline.

A.4 Functional literature bridge

Table 5: Background literature used to interpret the curated marker panel.

Region	Literature basis	Functional role carried into this paper
Telencephalon	Stednitz et al. (2018), Pandey et al. (2023), Reemst et al. (2023)	social orienting, learning, memory, and other forebrain regulatory functions
Optic tectum	Roeser and Baier (2003)	sensory integration, visuomotor transformation, orienting, and avoidance behavior
Proteomics article	Lubeckyj and Sun (2022)	region totals, named proteoform markers, duplicate summaries, and PTM counts used for the local re-analysis

A.5 Exact rebuild path

The package can be rebuilt locally with:

```
cd examples/codex-zebrafish-brain-paper-14rounds
./code/build_inputs.sh
./code/build_paper.sh
```

The analysis step should regenerate, at minimum:

- `results/summary_metrics.json`,

- `results/axis_summary.csv`,
- `results/group_consistency.csv`,
- the figure files under `figures/`.

The paper step should regenerate `paper/main.pdf`.

A.6 Marker-panel curation rule

The marker panel is intentionally conservative. A family enters the panel only if the source proteomics paper names it in the biological interpretation of telencephalon or optic tectum and if that interpretation can be connected to established zebrafish regional function in the cited background literature. The curated counts and family labels are materialized in `data/curated_evidence.json`, while the corresponding per-family summaries appear in `results/marker_bias.csv` and `results/group_consistency.csv`.

A.7 Leave-one-out robustness table

Table 6: Matched-region fraction after removing each marker family in turn.

Removed family	Remaining alignment
Pro-neuropeptide Y	0.9764
Synenkephalin	0.9764
Peptide YY-A	0.9764
Neurogranin a	0.9754
Synucleins total	0.9911
Reticulon	0.9758
Myosin	0.9625
Actin	0.9725
Troponin	0.9792

A.8 Claim-to-evidence ladder

Table 7: Main claims, supporting evidence, and corresponding limits.

Claim	Evidence	Main limit
Regions are sharply separated at the proteoform level	Jaccard overlap 0.0434, protein overlap 0.2151, unique fractions in Table 1	Derived from published counts rather than raw-feature reprocessing
Curated functional markers align with expected regional function	Axis-level alignment, exact test, family-level sign check, leave-one-out table	Claim is limited to the curated marker panel, not the full proteome
Pilot sensitivity is high enough for interpretable regional structure	Duplicate recovery fractions and single-run high-water-mark efficiency in Table 4	Still a pilot workflow rather than a large atlas
The paper improves auditability	Local code, figures, source manifest, and rebuild commands in this package	Does not replace the original wet-lab study